

int pressed=digitalRead(button);

22

23 if(pressed==HIGH){

24 buildUpTension();

25 delay(100);

26 int number=throwDice();

27 showNumber(number);

28 }

29 }

30

31 //THIS FUNCTION WILL LIGHTS THE LEDS FROM RIGHT TO LEFT THEN FROM LEFT TO

32 void buildUpTension(){

33 //from right to left

34 for(int i=first;i<=sixth;i++){

35 digitalWrite(i,HIGH);

36 digitalWrite(i-1,LOW);

37 delay(100);

38 }

39 //from left to right

40 for(int i=sixth;i>=first;i--){

41 digitalWrite(i,HIGH);

Output

Sketch uses 2138 bytes (6%) of program storage space. Maximum is 32756 bytes.



Search the web and Windows



```
29     digitalWrite(1-1,LOW);
30     delay(100);
31 }
32 //from left to right
33 for(int i=sixth;i>=first;i--){
34     digitalWrite(i,HIGH);
35     digitalWrite(i+1,LOW);
36     delay(100);
37 }
38 //loop to turn off all the leds
39 for(int i= first;i<=sixth;i++){
40     digitalWrite(i,LOW);
41 }
42 }
43
44 int throwDice() { //this function will generate a random number between 1 and 6
45     int randomNumber=random(1,7); //if we use random(1,6) the number 6 wont appear
46     Serial.print(randomNumber);
47     delay(500);
48     return randomNumber;
49 }
```

```
44 }  
23  
24 //THIS FUNCTION WILL LIGHTS THE LEDS FROM RIGHT TO LEFT THEN FROM LEFT TO RIGHT TO BUILD UP TENSION WAITING TO  
25 void buildUpTension(){  
26     //from right to left  
27     for(int i=first;i<=sixth;i++){  
28         digitalWrite(i,HIGH);  
29         digitalWrite(i-1,LOW);  
30         delay(100);  
31     }  
32     //from left to right  
33     for(int i=sixth;i>=first;i--){  
34         digitalWrite(i,HIGH);  
35         digitalWrite(i+1,LOW);  
36         delay(100);  
37     }  
38     //loop to turn off all the leds  
39     for(int i= first;i<=sixth;i++){  
40         digitalWrite(i,LOW);  
41     }  
42 }
```

```

22 }
23
24 //THIS FUNCTION WILL LIGHTS THE LEDS FROM RIGHT TO LEFT THEN FROM LEFT TO RIGHT TO BUILD UP TENSION WAITING TO
25 void buildUpTension(){
26   //from right to left
27   for(int i=first;i<=sixth;i++){
28     digitalWrite(i,HIGH);
29     digitalWrite(i-1,LOW);
30     delay(100);
31   }
32   //from left to right
33   for(int i=sixth;i>=first;i--){
34     digitalWrite(i,HIGH);
35     digitalWrite(i+1,LOW);
36     delay(100);
37   }
38   //loop to turn off all the leds
39   for(int i= first;i<=sixth;i++){
40     digitalWrite(i,LOW);
41   }
42 }

```

Output

Sketch uses 3128 bytes (6%) of program storage space. Maximum is 32768 bytes.

10:31 AM | Arduino IDE 2.3.10 | 10:31 AM



Search the web and Windows



10:31 AM 10:31 AM

```
22 }
23
24 //THIS FUNCTION WILL LIGHTS THE LEDS FROM RIGHT TO LEFT THEN FROM LEFT TO RIGHT TO BUILD UP TENSION WAITING TO
25 void buildUpTension(){
26     //from right to left
27     for(int i=first;i<=sixth;i++){
28         digitalWrite(i,HIGH);
29         digitalWrite(i-1,LOW);
30         delay(100);
31     }
32     //from left to right
33     for(int i=sixth;i>=first;i--){
34         digitalWrite(i,HIGH);
35         digitalWrite(i+1,LOW);
36         delay(100);
37     }
38     //loop to turn off all the leds
39     for(int i= first;i<=sixth;i++){
40         digitalWrite(i,LOW);
41     }
42 }
```

Activate Windows

Go to Settings to activate Windows.

Output

Sketch uses 3120 bytes (6%) of program storage space. Maximum is 32768 bytes.

1x11 Duino - Arduino Uno (2016 not connected)



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ENG

5:28 PM
11/11/2023

The screenshot shows a presentation slide titled "Dice Challenge" with the subtitle "How to think before Connecting". The slide lists three tasks:

3. Build a ~~void buildUpTension()~~ function to light LEDs from left to right, and back to build up tension while waiting for the dice to be thrown from left to right and from right to left.
4. Build a ~~int throwDice()~~ function to get a random number in the range [1,6] return that ~~getRandom()~~, and we can print it in the window of the serial monitor.
5. Build a ~~void showNumber(int number)~~ function to turn on LEDs depending on the random number obtained by throwing the dice.

On the right side of the slide, there is an illustration of a man in a green shirt thinking, with his hand on his chin. The slide is part of a presentation with 15 slides, and the current slide is slide 8 of 15.

[illegible]

12 - Challenge 1: Dice Application - PowerPoint

File Home Insert Design Layout References Animations Slide Show Review View Developer Assist Tell me... Support Sign Out

Outline New slide Table of contents Shapes Arrange Quick styles Editing Create a PDF

Outline Slide Show View

Slide 7 of 15

Dice Challenge

How to think before Connecting

Question & Answer

1. Define 6 consecutive digital pins for the LEDs and pin for the button switch, and a variable "pressed" to check state of button switch (0: pressed = 0).

2. In void setup():

- a) set all LED pins to OUTPUT
- b) set button pin to INPUT
- c) initialize random seed by noise from analog pin 0 (should be unconnected)

`randomSeed(analogRead(0));`

etc.

Sketch - Arduino IDE 2.3.10

File Edit Sketch Tools Help

Arduino IDE

Sketch: j071a

```
1 int first=7;
2 int second=8;
3 int third=9;
4 int fourth=10;
5 int fifth=11;
6 int sixth=12;
7 int button=4;
8
9 void setup() {
10   // put your setup code here, to run once:
11   pinMode(button,INPUT);
12   for(int i=first;i<=sixth;i++){
13     pinMode(i,OUTPUT);
14   }
15   Serial.begin(9600);
16 }
17
18 void loop() {
19   // put your main code here, to run repeatedly:
20   for(int i =first;i<=sixth;i++){
```

Sketch uses 3138 bytes (6%) of program storage space. Maximum is 4096 bytes.

Dice Challenge

How to think before Connecting

- We also have a call to `randomSeed` in the setup method. If this was not there, every time we reset the board, we would end up with the same sequence of dice throws. (2 - 4 - 1 - 5 - 3 - 6).
- Load the sketch in the Board from the IDE. Pressing the button will start rolling the dice.



Dice Challenge

How to think before Connecting

- Connect the LEDs as shown in schematic. The pins for LEDs are {7,8,9,10,11,12}. The Button (SwitchPin) is connected to pin 4.
- There are a few nice touches that make the dice behave in a similar way to real dice. For example, as the dice rolls, the number changes but gradually slows. The length of time that the dice rolls is also random.
- We now have six LEDs to initialize in the setup method, so it is worth putting them in an array and looping over the array to initialize each pin.



Layout 14 ☐ Test Direction ☐ Align Left ☐ Convert to SmartArt

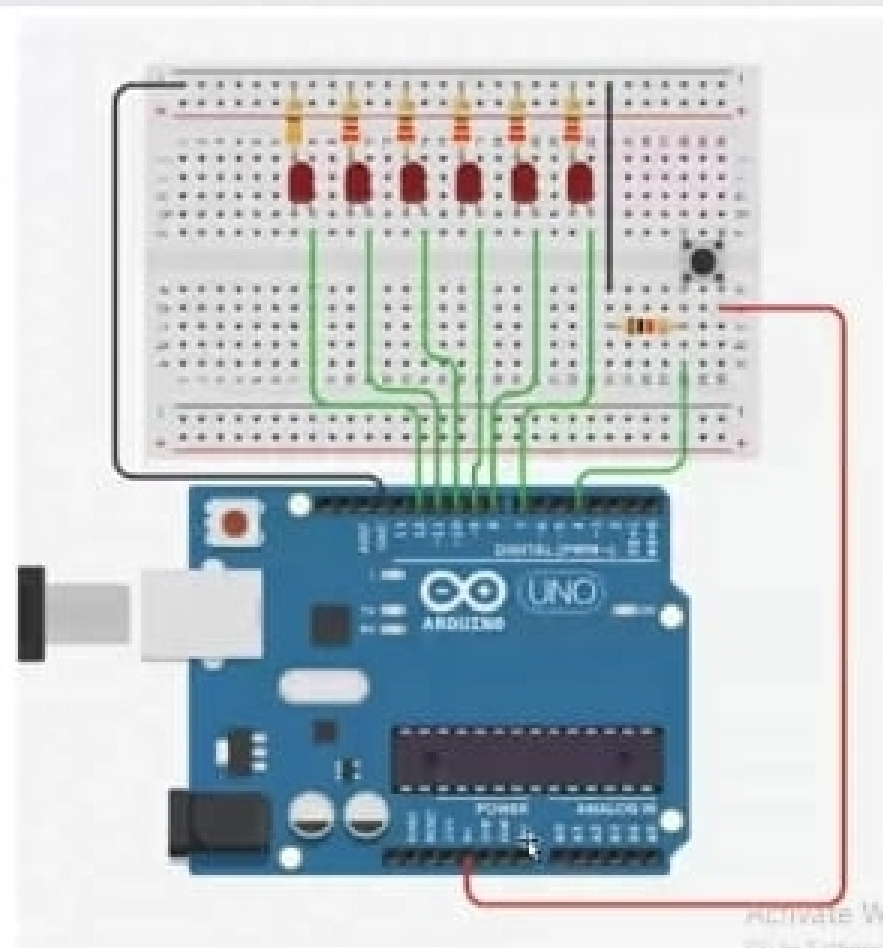
New Slide Section

Slide Paragraph

Arrange Colors Styles Shape Fill Shape Outline Shape Effects

Print Revert Select Create a PDF Active Acrobat

Board connection



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Sketch_Jul27b.ino

Arduino Uno

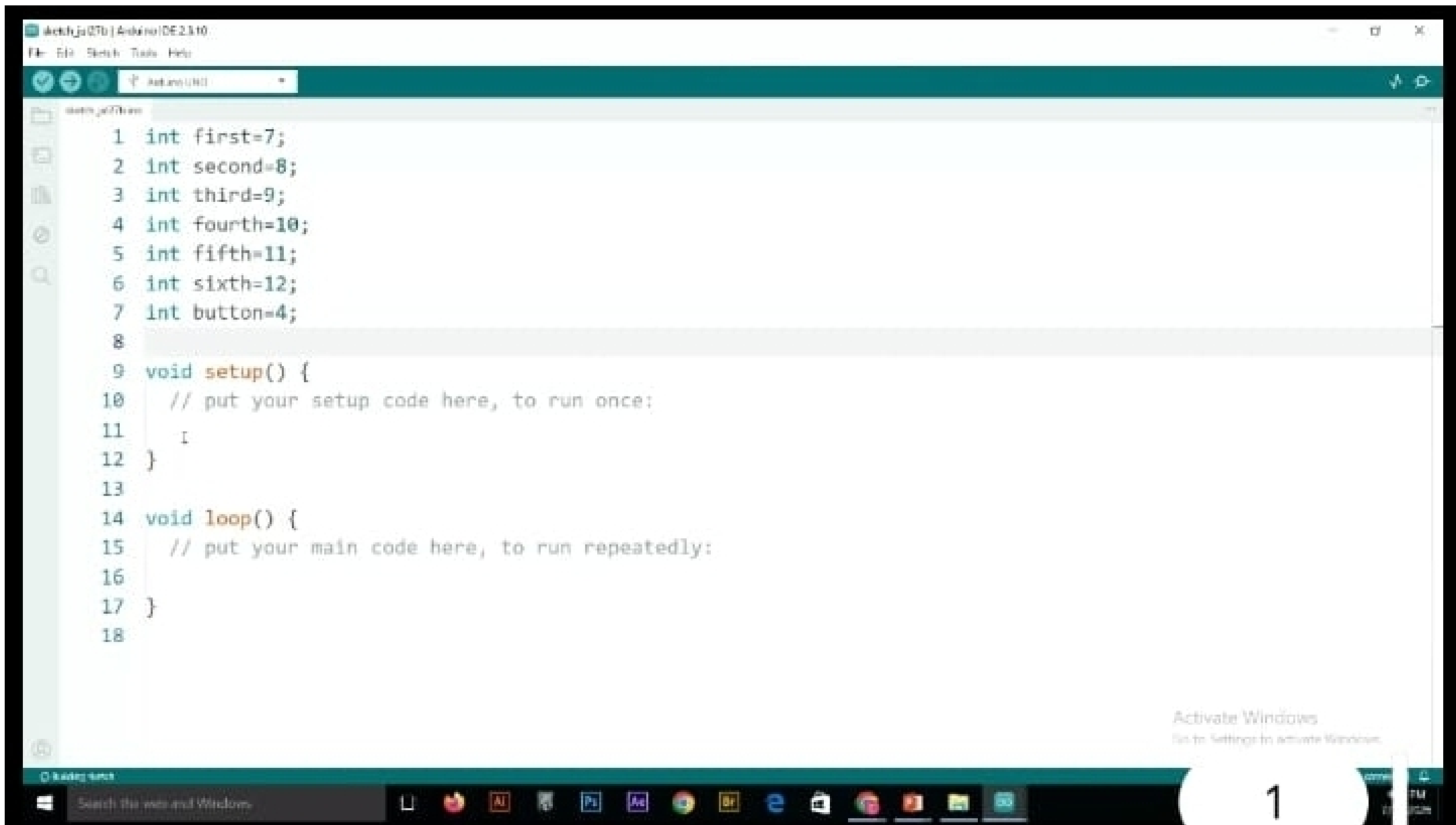
Upload

```
6 int sixth=12;
7 int button=4;
8
9 void setup() {
10   // put your setup code here, to run once:
11   pinMode(button,INPUT);
12   for(int i=first;i<=sixth;i++){
13     pinMode(i,OUTPUT);
14   }
15   Serial.begin(9600);
16 }
17
18 void loop() {
19   // put your main code here, to run repeatedly:
20   for(int i =first;i<=sixth;i++){
21     digitalWrite(i,HIGH);
22   }
23   int buttonstate=digitalRead(button);
24   Serial.print((buttonstate==HIGH)? "BUTTON PRESSED":"BUTTON NO
25   delay(500);
26 }
```

Output

Sketch uses 2118 bytes (6%) of program storage space. Maximum is 32768 bytes.





Layout
Insert
Section

Font
Paragraph

Text Direction
Align Text
Convert to SmartArt

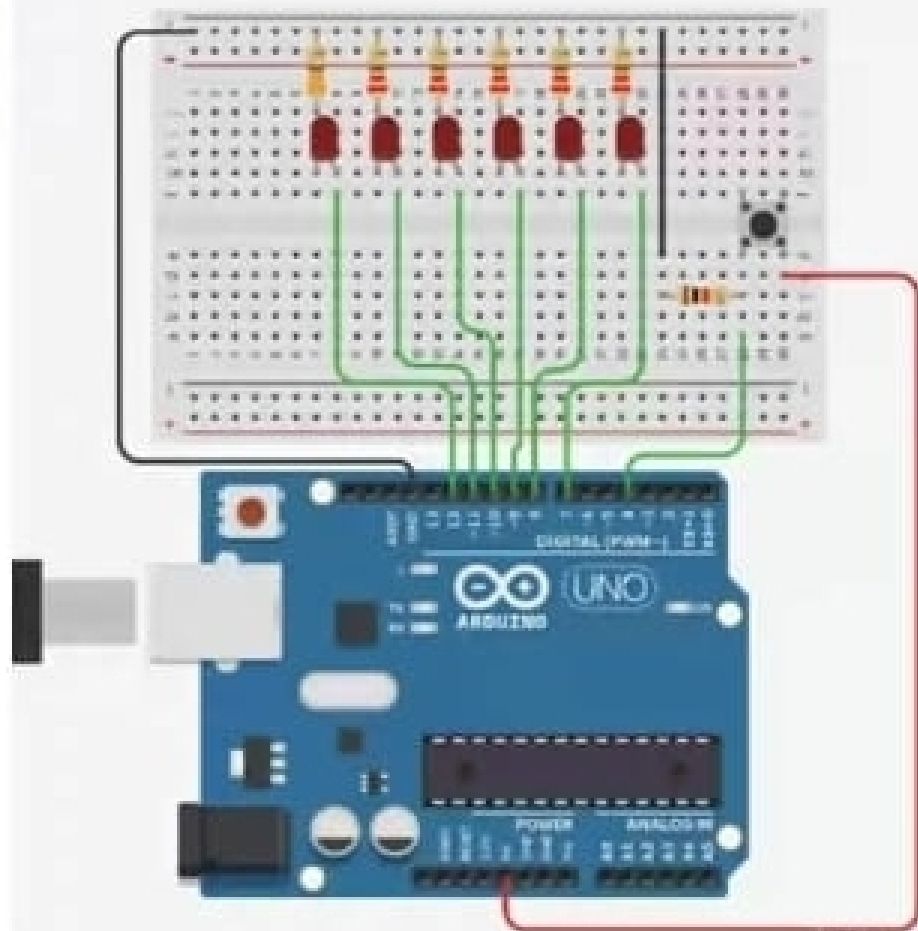
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Drawing

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Quick Styles
Shape Effects

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Replace
Select

Editing
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Board connection



dot

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Slides Comments

Dice Challenge

Required Elements:



Arduino



X7 Resistor 330 Ω



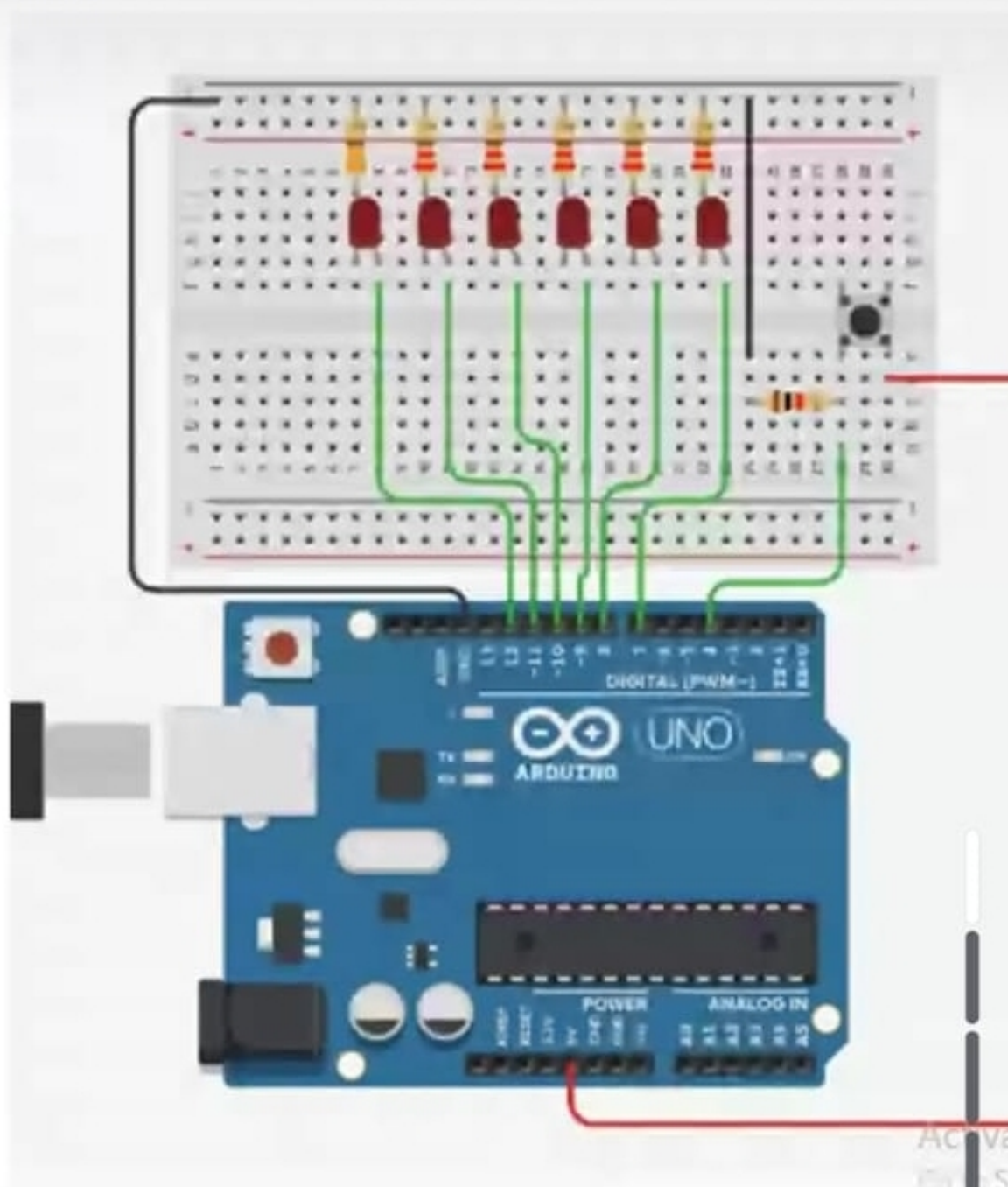
Red LED



Push Button

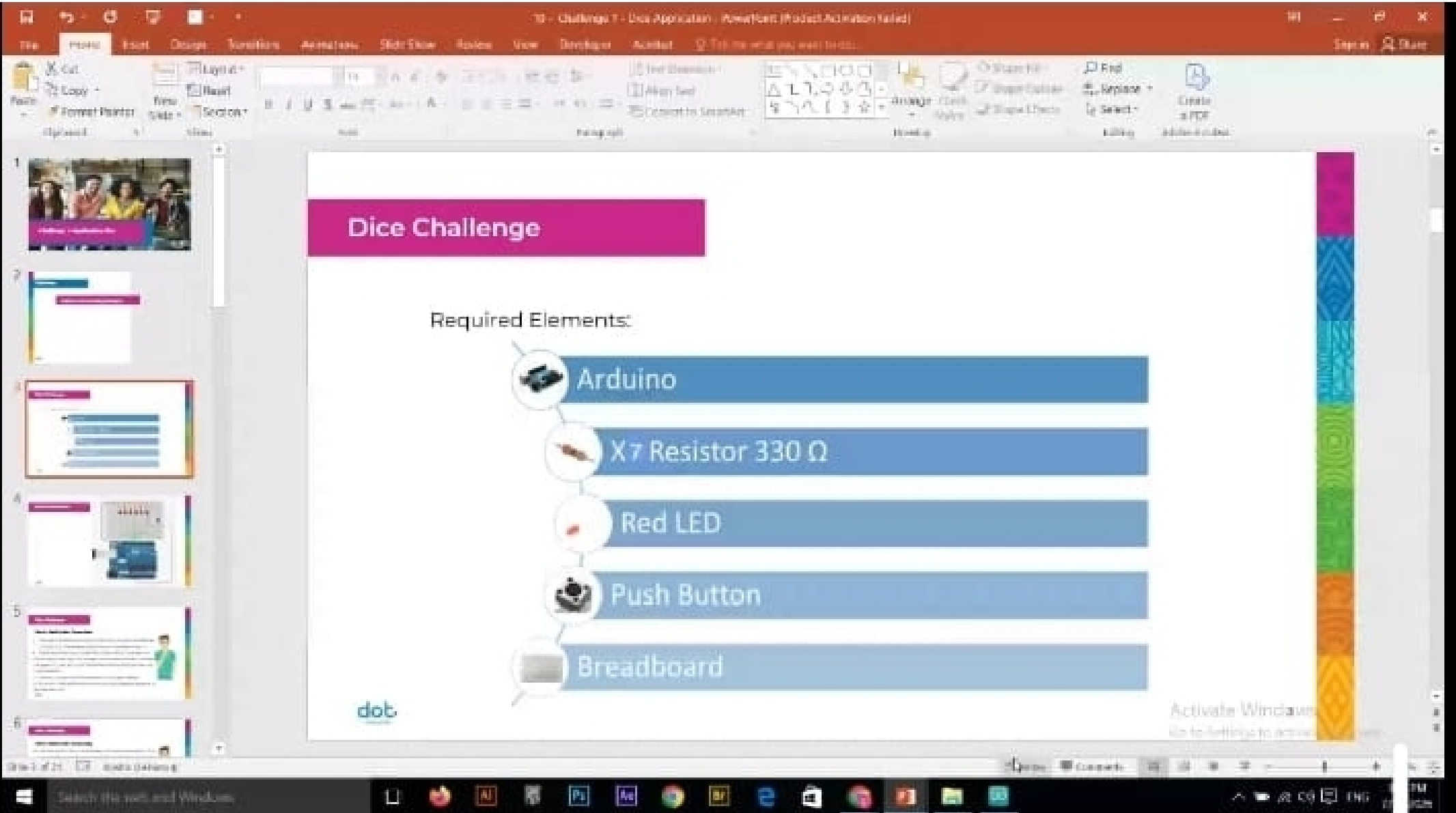


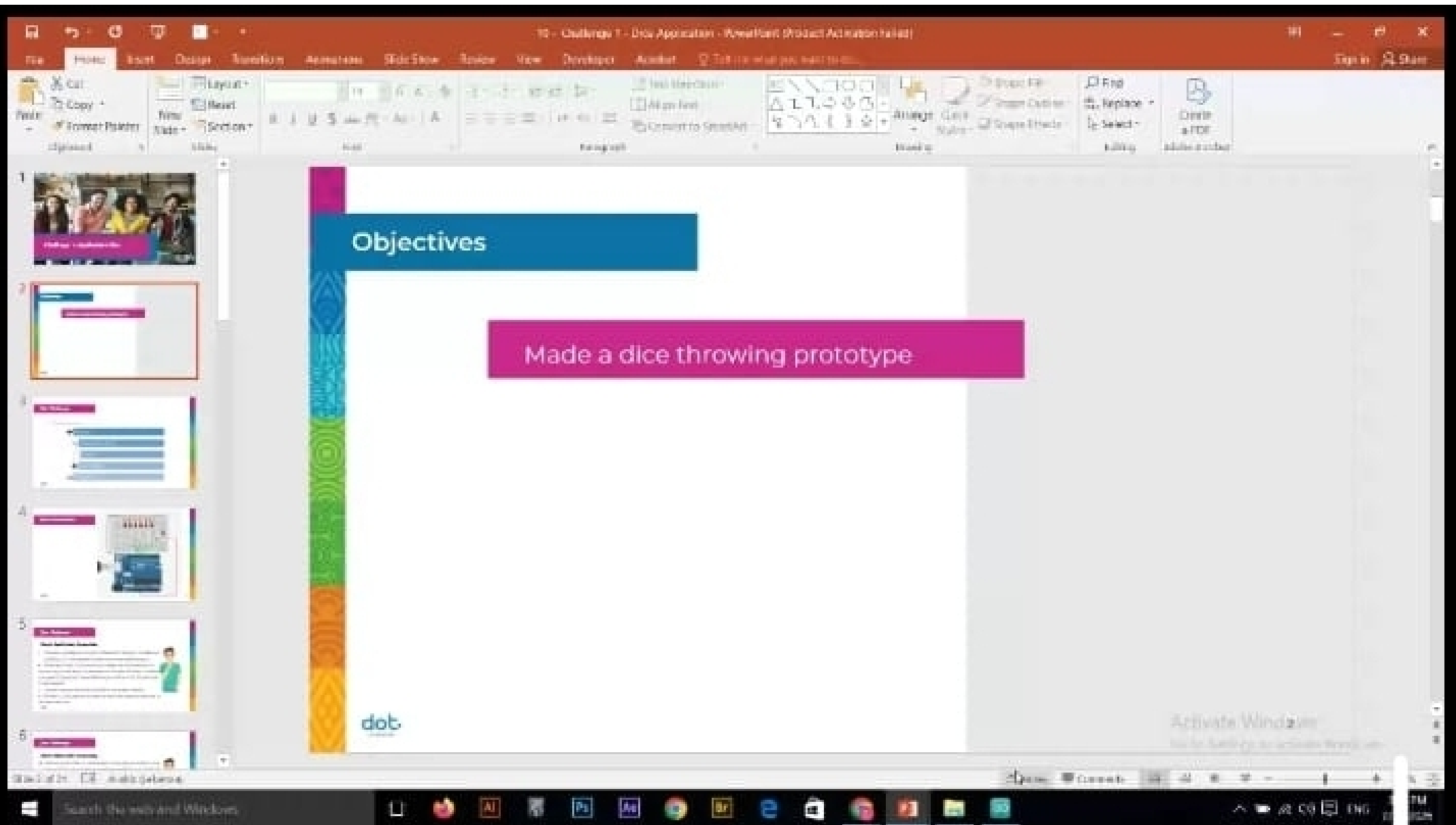
Breadboard



States Comments



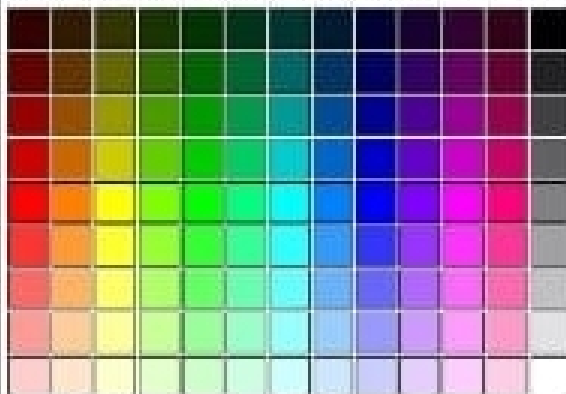






RGB color codes chart

Hover with cursor on color to get the hex and RGB color codes below:



Hex: #

Red:

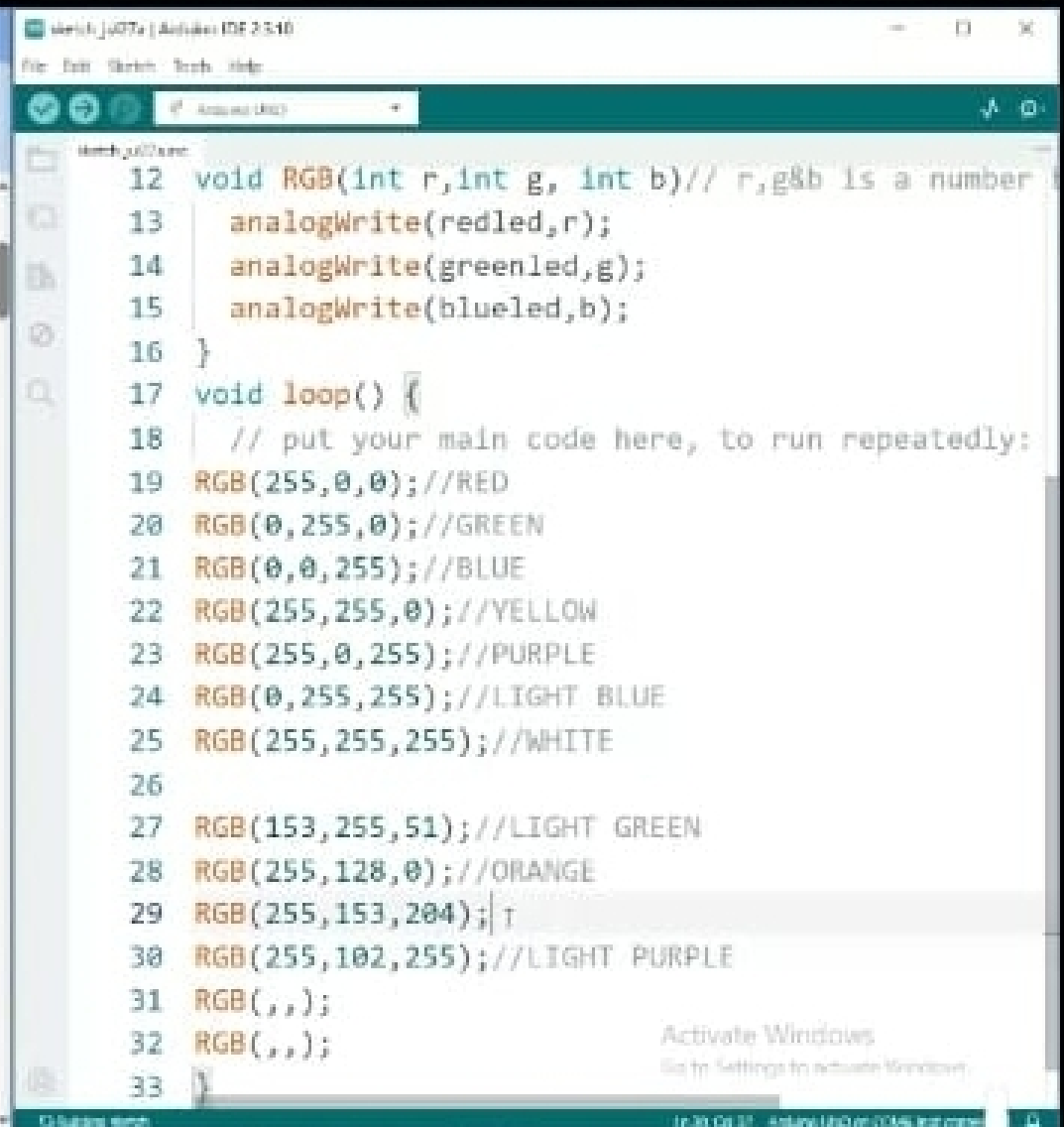
Green:

Blue:



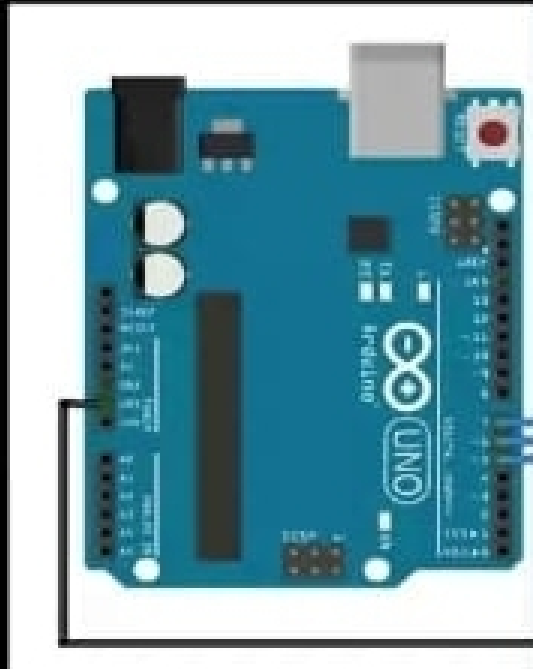
RGB color space

RGB color space or RGB color system, constructs all the colors from the combination of the Red,



Photos

Photos app



Sketch: ju07a | Arduino IDE 2.5.10

File Edit Sketch Tools Help

Arduino Uno

Sketch: ju07a.kicad

```
14   analogWrite(greenled,g);
15   analogWrite(blueled,b);
16 }
17 void loop() {
18   // put your main code here, to run repeatedly:
19   RGB(255,0,0); //RED
20   RGB(0,255,0); //GREEN
21   RGB(0,0,255); //BLUE
22   RGB(255,255,0); //YELLOW
23   RGB(255,0,255); //PURPLE
24   RGB(0,255,255); //LIGHT BLUE
25   RGB(255,255,255); //WHITE
26
27   RGB(153,255,51); //LIGHT GREEN
28   RGB(255,128,0); //ORANGE
29   RGB(255,153,204); //LIGHT PINK
30   RGB(255,102,255); //LIGHT PURPLE
31
32 }
33
```

Activate Windows

Go to Settings to activate Windows.



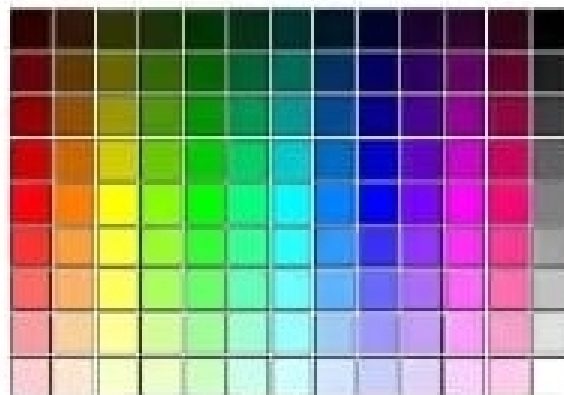
Search the web and Windows



Color picker interface showing RGB values: R 255, G 255, B 0. The hex code is #80FF00.

RGB color codes chart

Hover with cursor on color to get the hex and RGB color codes below:



Hex: # FF99CC
Red: 255
Green: 153
Blue: 204



RGB color space

RGB color space or RGB color system, constructs all the colors from the

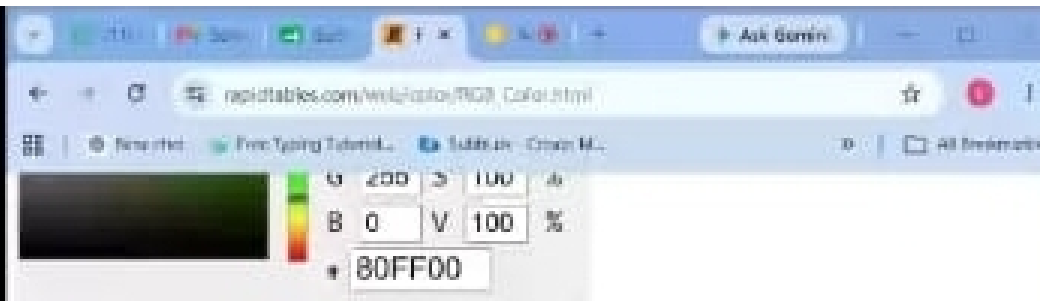
- 3rd Grade
- 4th Grade
- 5th Grade
- 6th Grade
- 7th Grade
- 8th Grade



WEB COLORS

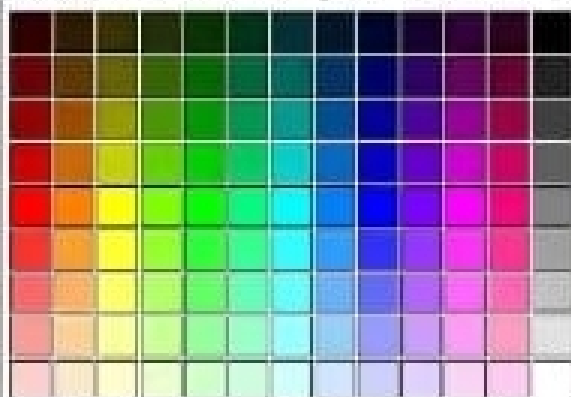
- RGB color
- Color picker
- Color scheme
- Color tester
- Color wheel
- HTML color codes
- Web safe colors
- Blue color

Activate Windows
Go to Settings to activate Windows.



RGB color codes chart

Hover with cursor on color to get the hex and RGB color codes below:



Hex: # FFFFFFFF

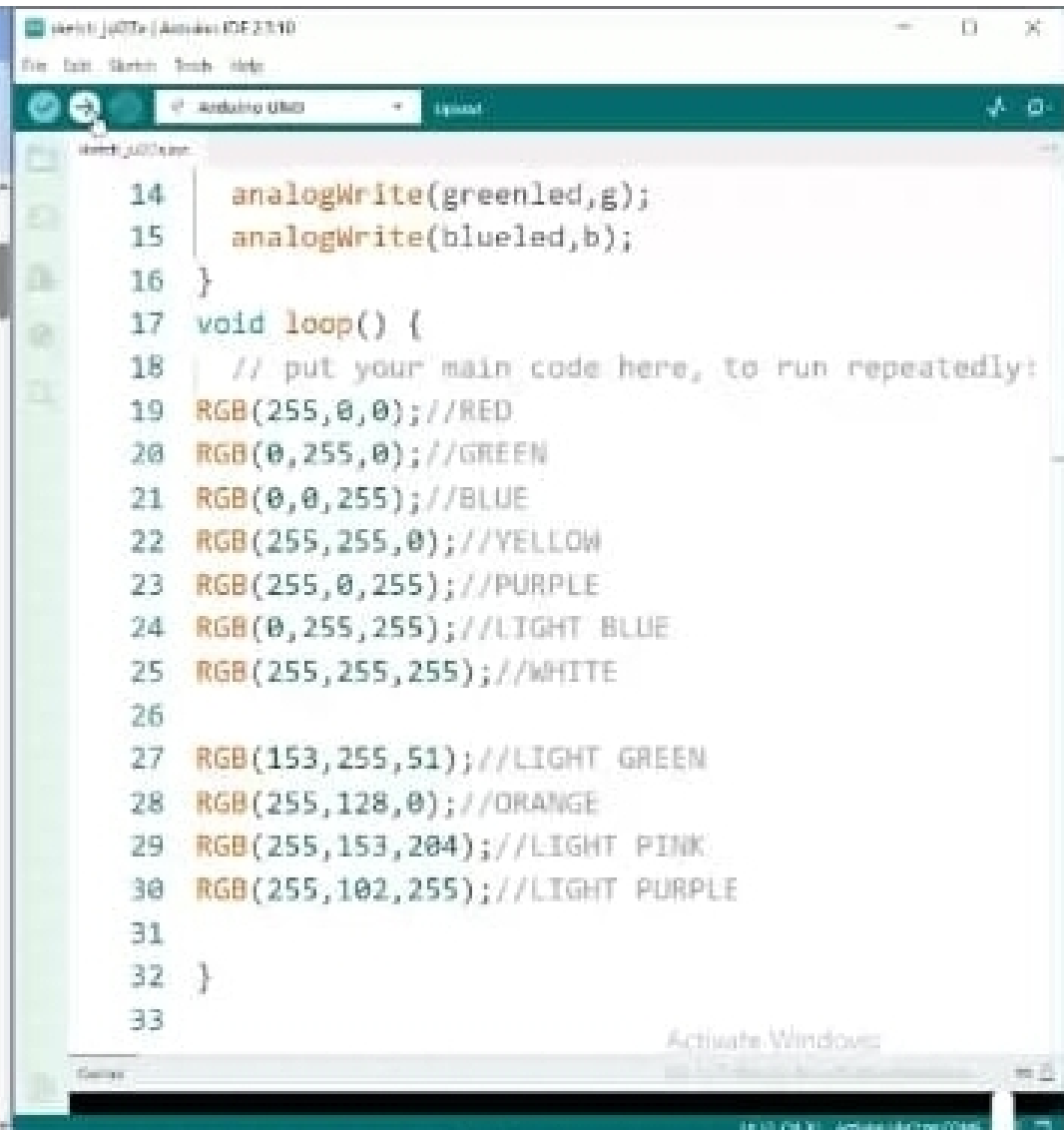
Red: 255

Green: 255

Blue: 255

RGB color space

RGB color space or RGB color system, constructs all the colors from the combination of the Red,



Activate Windows

Go to Settings to activate Windows.

10:10 AM - 10/10/2023

10/10/2023 10:10 AM

```
1 int redled=9; //ALL THE LEDS SHOULD BE CONNECTED ON PWM PINS(3-5-6-9-10-11)
2 int greenled=10;
3 int blueled=11;
4
5 void setup() {
6   // put your setup code here, to run once:
7   pinMode(redled,OUTPUT);
8   pinMode(greenled,OUTPUT);
9   pinMode(blueled,OUTPUT);
10 }
11
12 void RGB(int r,int g, int b)// r,g&b is a number between 0 and 255{
13   analogWrite(redled,r);
14   analogWrite(greenled,g);
15   analogWrite(blueled,b);
16 }
17 void loop() {
18   // put your main code here, to run repeatedly:
19
20 }
21
```

Activate Windows

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Help

CPU 15.0% RAM 26% Arduino IDE on COM6 (not connected)



Search the web and Windows



^ > < < > ENG



7:01 PM

sketch_jul27a.ino

```
1 int redled=9; //ALL THE LEDS SHOULD BE CONNECTED ON PWM PINS(3-5-6-9-10-11)
2 int greenled=10;
3 int blueled=11;
4
5 void setup() {
6   // put your setup code here, to run once:
7   pinMode(redled,OUTPUT);
8   pinMode(greenled,OUTPUT);
9   pinMode(blueled,OUTPUT);
10 }
11
12 void RGB(int r,int g, int b){
13   analogWrite(redled,r); // r is a number between 0 and 255
14 }
15 void loop() {
16   // put your main code here, to run repeatedly:
17
18 }
19
```

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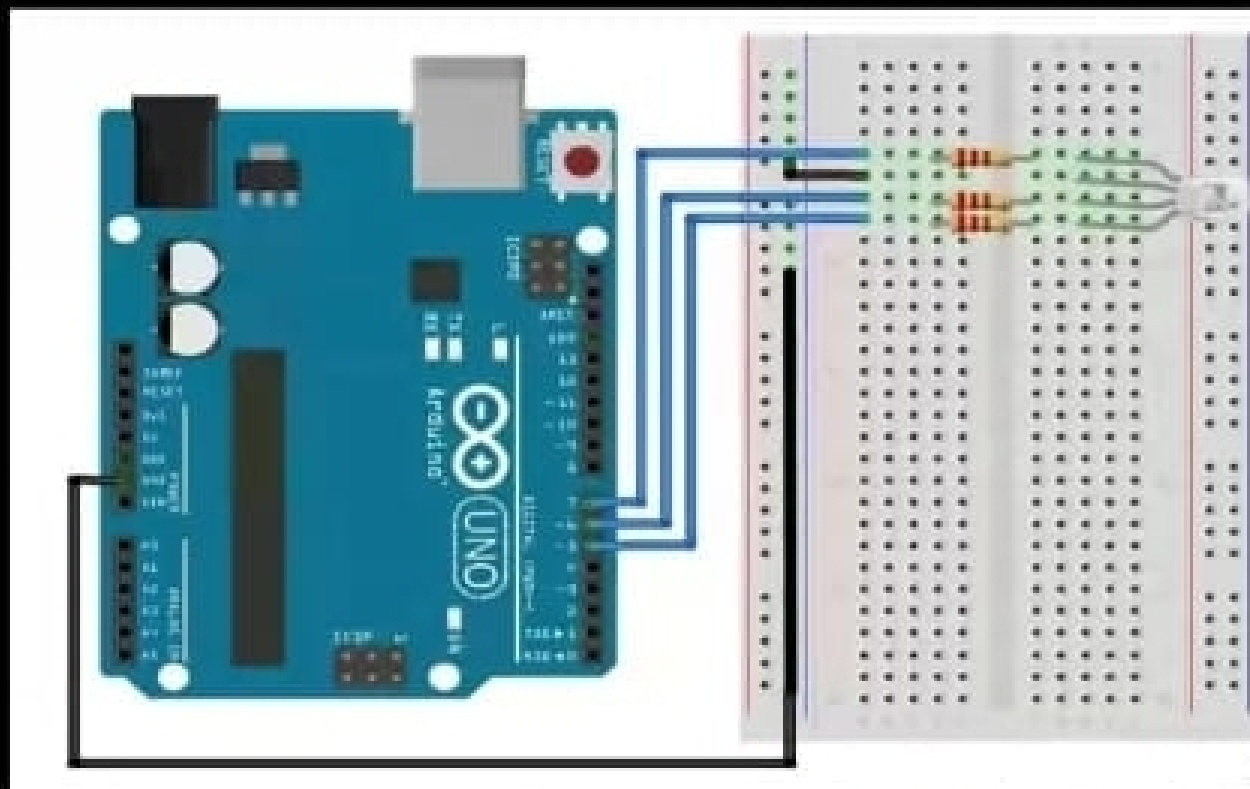
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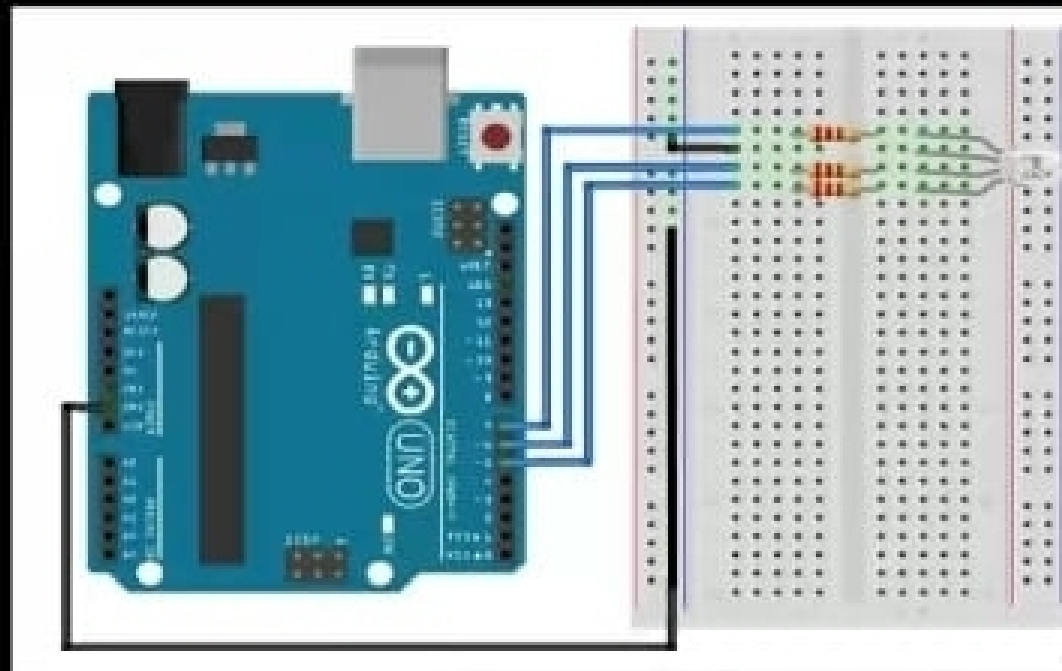
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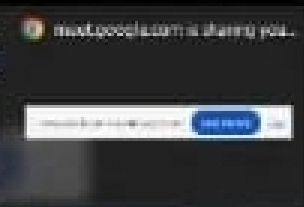


Photos

Photos App



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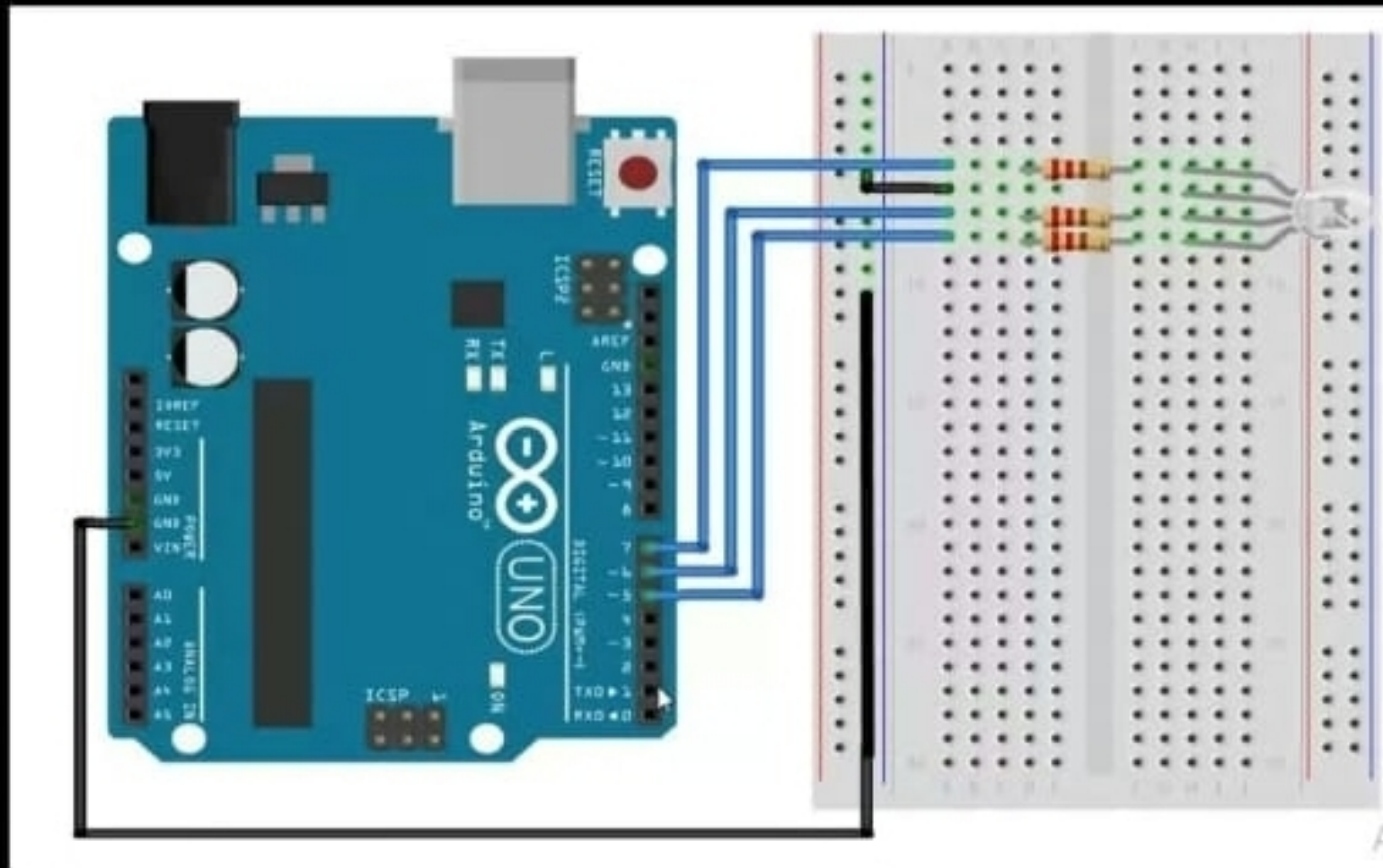
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Search the web and Windows



1:15 PM
7/1/2020



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